

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Easy

Question
 $(2x^3 + 3x)(x^3 - 2x)$

Which of the following is equivalent to the expression above?

- Answer**
- A. $x^3 + 5x$
- B. $3x^3 + x$
- C. $2x^6 - x^4 - 6x^2$
- D. $3x^6 - x^4 - 6x^2$

Correct Answer: C

Rationale
Choice C is correct. Using the distributive property to multiply the terms in the parentheses yields $(2x^3 \cdot x^3) + (2x^3 \cdot -2x) + (3x \cdot x^3) + (3x \cdot -2x)$, which is equivalent to $2x^6 - 4x^4 + 3x^4 - 6x^2$. Combining like terms results in $2x^6 - x^4 - 6x^2$.

Choices A and D are incorrect and may result from conceptual errors when multiplying the terms in the given expression. Choice B is incorrect and may result from adding, instead of multiplying, $(2x^3 + 3x)$ and $(x^3 - 2x)$.